

BUSINESS PROCESS FLOW: SMARTLOGI SYSTEM

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Project: SmartLogi - AI-Powered Logistics Management

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1. EXECUTIVE SUMMARY

This document outlines the end-to-end business workflows of the SmartLogi platform. The system integrates **Agentic AI** for decision-making and **Retrieval-Augmented Generation (RAG)** for policy compliance and document processing. These flows are designed to minimize manual intervention and maximize "Last-Mile" delivery efficiency.

2. CORE BUSINESS WORKFLOWS

2.1 Order Ingestion & AI Validation

This is the entry point where customer requests are converted into actionable logistics tasks.

- Request Input:** Customers or Warehouse Staff submit delivery requests via the Web Portal or Mobile API.
- AI Data Validation:** An autonomous **Validation Agent** checks the integrity of the shipping address, contact details, and cargo type.
- RAG Policy Check:** The system cross-references the order against internal shipping policies (stored in the Knowledge Base) to ensure the items are shippable (e.g., checking hazardous material restrictions).
- Confirmation:** If valid, the order status transitions to "**Ready for Dispatch.**"

2.2 Intelligent Route Optimization (AI-Driven)

The core value proposition of SmartLogi, where the **Agentic AI** creates the most efficient delivery path.

- Grouping:** Orders are automatically clustered based on geographical proximity and delivery deadlines.

- **Environmental Analysis:** The AI Agent retrieves real-time data regarding traffic congestion, weather conditions, and vehicle availability.
- **Optimization Algorithm:** AI calculates the "Shortest Path" while balancing vehicle load capacity and fuel efficiency.
- **Automatic Dispatch:** The system assigns the optimized route to the nearest available **Driver** and pushes the data to their Mobile App.

2.3 Field Execution & Real-Time Tracking

The physical movement of goods handled by the **Driver** actor.

1. **Mobile Notification:** The Driver receives a push notification containing the optimized route and turn-by-turn navigation.
2. **QR Verification:** Upon arrival at the warehouse, the Driver scans a QR code to confirm the "Pick-up" of specific packages.
3. **Execution:** The Driver follows the AI-suggested route, providing real-time GPS pings back to the **Admin Dashboard**.
4. **Proof of Delivery (PoD):** Upon arrival at the destination, the Driver captures a digital signature or a photo as proof.
5. **Completion:** The system updates the status to "**Delivered**," triggering the invoicing process.

3. VISUAL PROCESS MAP (MERMAID)

This diagram visualizes the logic flow for developers and stakeholders.

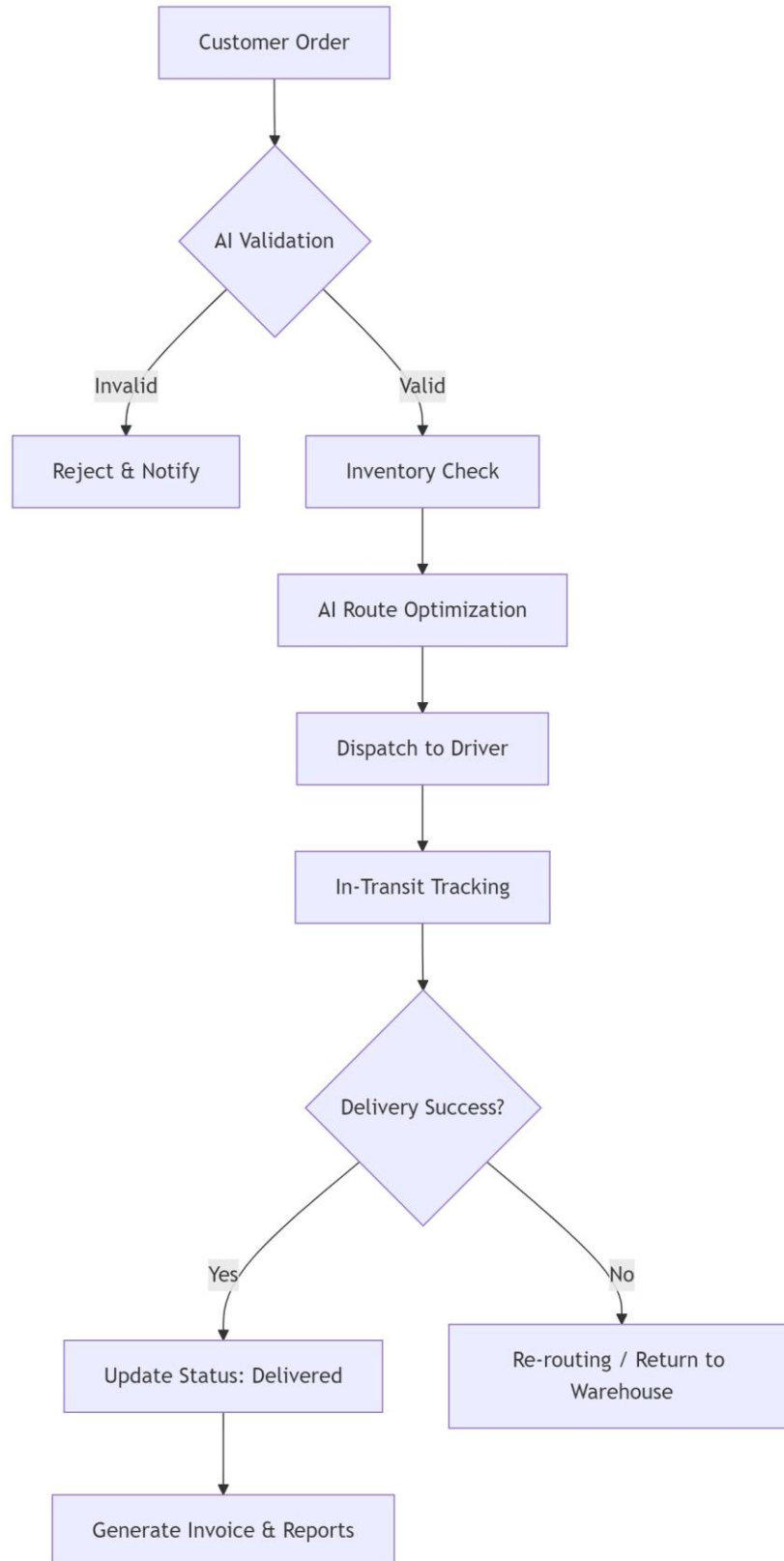


Figure 1 Visual process map

4. EXCEPTION & RISK MANAGEMENT

- **Route Disruption:** In the event of an accident or roadblock, the **AI Agent** detects the GPS delay and automatically recalculates an alternative route in real-time.
- **Failed Delivery Attempt:** If a customer is unavailable, the system logs a "Failed Attempt" and automatically reschedules the delivery for the next available window based on current route density.

5. STAKEHOLDER RESPONSIBILITIES

- **Logistics Manager:** Oversees the "Macro" flow and handles exceptions flagged by the AI.
- **AI Agent:** Performs the "Micro" decisions (Validation, Routing, and Notifications).
- **Driver:** Executes the physical delivery and provides "Ground-Truth" data updates via the app.